

Multiple Choice (10 marks)

1. In Python 3, what is the output of `['Hi!'] * 4`?

`'Hi!', 'Hi!', 'Hi!', 'Hi!'`

`'Hi!' * 4`

- (a) Error
 - (b) None of the above.
2. What is the output of `"abc"[-1]` in Python 3?
- (a) Error
 - (b) a
 - (c) b
 - (d) c
3. Which of the following best describes the primary way in which a distributed denial-ofservice (D D o S) attack differs from a denial-ofservice (D o S) attack?
- (a) The goal of the attack
 - (b) The number of computers being attacked
 - (c) The number of computers launching the attack
 - (d) The time period in which the attack occurs
4. Which of the following is the hexadecimal representation of the decimal number 231_{10} ?
- (a) 17_{16}
 - (b) E_4_{16}
 - (c) E_7_{16}
 - (d) F_4_{16}
5. Let `l = [1,2,2,3,4]`. In Python 3, what is a possible output of `set(l)`?
- (a) `{1}`
 - (b) `{1,2,2,3,4}`
 - (c) `{1,2,3,4}`
 - (d) `{4,3,2,2,1}`

True / False (10 marks)

Answer True or False

6. True or False: Confidentiality is compromised when a user can access a system without proper authentication.
7. True or False: The function $O(N^{1/4})$ grows slower than $O(N^{1/N})$ as N becomes very large.
8. True or False: The boolean expression $a[i] == \max \parallel \neg(\max \neq a[i])$ simplifies to $a[i] == \max$, indicating equality with $\max$.
9. True or False: In Python 3, the operator `'/'` is used for floor division.
10. True or False: The method `'sort()'` can be used directly on strings to sort their characters in Python.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how you would approach modifying the specification for a method that searches an array for values larger than a given item, particularly addressing the handling of multiple indices.
12. Discuss the potential benefits of upgrading a customer service system in a company. Which benefit might be the least likely to improve, and why?

End of Paper